Ian Keilman

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SUMMARY

Software Engineer with full-time contract experience and 3 internships building backend/database systems, internal tools, and data-heavy applications. Strong with Python, TypeScript, C/C++, SQL, PostgreSQL, React/Next.js, and Linux.

EDUCATION

University of California, Santa Cruz **2026**
Bachelor's Degree, Computer Science and Engineering Focus in networking & digital technology

EXPERIENCE

Software Engineer | Contract, Project Lead
American Soil Testing & Engineering Inc. - Alamo, CA (Remote) | August 2026 - Present

• Led a small development team to deliver a custom reporting application, solely owning UI/UX design and system architecture while translating evolving client requirements using TypeScript, React, Node.js, and Electron.

• Built reusable report templates, autofill workflows, and a PDF/DOCX ingestion pipeline using PDF.js and Tesseract OCR to support editing legacy documents containing text, images, tables, and metadata.

• Engineered schema-validated offline persistence with atomic file writes, IndexedDB, and filesystem synchronization, enabling offline editing and file sharing through a Google Drive–synced folder without a custom cloud backend.

• Developed collision-aware document layout, cross-page reflow, and PDF export; expanded the regression suite to 355 automated tests across 26 test files.

Software Engineering Intern | Backend & Database
Zee Foods, LLC. - Fremont, CA (Remote) | May 2026 - August 2026

• Built backend/database features for Polaris, a full-stack internal operations platform replacing B2B SaaS workflows

• Created order management features with PostgreSQL schemas, Drizzle, CRUD, server actions, and SKU-linked records

• Added RLS/CASL authorization, real-time UI updates, integration tests, and end-to-end validation

Software Engineering Intern | Robotics & Automation
Bishop-Wisecarver - Pittsburg, CA | February 2026 - May 2026

• Built a local torque-spec lookup tool to map product/part numbers to fastener lists and reduce manual lookup time

• Programmed and calibrated robotic arm routines for automated scanning workflows, validating motion behavior

• Validated computer vision training data in CVAT through labeling QA, spot reviews, and consistency audits

Software Engineering Intern | Backend & Data Automation
Grand Petroleum Inc. - Pleasant Hill, CA (Remote) | September 2024 - August 2025

• Built an Excel/VBA automation suite that converted raw multi-store exports into standardized sales/margin reports

• Added automated validation and exception flagging for missing data, outliers, and pricing/cost/profit mismatches

• Designed one-click workflows for non-technical users, reducing manual spreadsheet review time

PROJECTS

sorter.ly | Next.js, React, TypeScript, SQLite, Drizzle ORM
• Local-first customizable collection manager with dynamic fields, typed records, and private SQLite-backed storage
• Implemented record CRUD, typed sorting and filtering, and CSV import/export in a full-stack web application

HedgeSlice | C++, STL, GoogleTest
• Implementing a price-time priority limit order book with limit/market orders, cancellations, and trade execution logs
• Building matching logic with ordered maps, queues, and hash tables; validating edge cases with unit tests

SKILLS

Languages: C/C++, Python, Java, JavaScript/TypeScript, SQL, HTML/CSS, VBA | **Web Development:** React.js, Next.js
Backend/Data: REST APIs, authentication, CRUD, data validation, PostgreSQL, SQLite, Drizzle ORM, SQLAlchemy
Tools: Git, Linux, Bash, Docker, Node.js, Excel, GitHub Actions | **Machine Learning/Computer Vision:** CVAT, dataset QA